

INDIA'S LUNAR MISSION CHANDRAYAAN-1

India's lunar spacecraft Chandrayaan has already started releasing images of the moon surface. Let's review its journey, payloads and mission objectives

■ DR S.S. VERMA

India's first unmanned moon mission from Sriharikota (Nellore district in Andhra Pradesh) with orbiting spacecraft Chandrayaan-1 (moon craft) tak-

ing the moon impact probe (MIP) has been launched successfully. The MIP having Indian tricolour painted on all its sides landed on the moon surface after having flown 386,000 km from the earth, making India the fourth country on the earth to land a probe on the moon. Indian scientists involved in particular and all the public in general are very proud of this scientific and technological endeavor.

According to Dr G. Madhavan Nair, chief of Indian Space Research Organisation (ISRO), "We have given moon to India." Every concerned Indian has traveled all the way to the moon virtually along the MIP. The MIP has already sent images with high resolution of the moon for analysis.

The United States,

the former Soviet Union and the European Space Agency comprising 17 countries already have their flags on the moon and Chandrayaan-1 is the world's 68th mission. Chandrayaan-1 was assembled to have eleven payloads. The spacecraft being read-

ied at another building was moved to the vehicle building to couple Chandrayaan-1 with the launch vehicle and with all the payloads it was then moved to the launch pad. The successful flight of MIP is supposed to prepare for the second

Indian moon mission, Chandrayaan-2, which will carry a Russian rover and a lander slated for lift-off between 2010 and 2012.

Let's review Chandrayaan's journey, payloads and mission objectives.

Space probe

A space probe is a scientific space exploration mission in which a robotic spacecraft leaves the gravity well of the earth and approaches the moon or enters interplanetary or interstellar space; approximately twenty are currently existent. The space agencies of the USSR (now Russia and Ukraine), the United States, the European Union, Japan and China have in the aggregate launched probes to several planets and moons of the solar system as well



Fig. 1: Launch of Chandrayaan-1